

EBN111 SEMESTER TEST/ TOETS 2 (22 APR 2016)
ANSWER SHEET / ANTWOORDBLAD

Surname: Van:	Memo.	Student number: Studentenommer:						
Initials: Voorletters:		Group: Groep:	ENG	ENG (Mech)	AFR			

Q1	13	Q2	22	Q3	11	Tot:	46	/45
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QUESTION 1 - VRAAG 1 (13)				
1.1	$a =$	9	$b =$	5
			$c =$	-1
				$d =$
1.2	$\begin{pmatrix} 0.5 & -0.25 & 0 \\ -0.25 & 0.45 & -0.2 \\ 0 & -0.2 & 0.7 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \\ 2 \end{pmatrix}$			
1.3	$a) V_S =$		6.7 V	
			$b) V_{CE} =$	
			14.04 V	
1.4	$i_0 =$			
	0.5 mA			

QUESTION 2 - VRAAG 2(22)	
2.1	$V_T =$
	18V
2.2	$V_S =$
	20V
	$I_S =$
	4A
2.3	$I_X =$
	3.3 mA
2.4	$V_Y =$
	-6V
2.5	$V_{TH} =$
	10V
	$R_{TH} =$
	41.52
2.6	$I_N =$
	4A
	$R_N =$
	252
2.7	$a) P_{max} =$
	0.125W
	$b) P_L =$
	0.5W

VRAAG 3 – QUESTION 3 (11)

3.1	a) $\frac{v_o}{v_i} = 11$ ✓✓✓ b) $i_x = -0.6 \text{ mA} = -600 \mu\text{A}$ ✓✓✓
3.2	a) Inverting amplifier ✓✓ b) $V_0 = -50 \text{ mV}$ ✓✓✓ 0.05 V ✓✓✓
3.3	$v_s = 10.5 \text{ mV}$ ✓✓✓ $= 0.0105 \text{ V}$ ✓✓✓
3.4	$v_o = -2 \text{ mV}$ ✓✓✓